

SIMATS ENGINEERING

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0405

CO 3 – ASSESSMENT TOOL 2: DATA INTERPRETATION

Prepared Solutions for Assigned Questions 1–5

Course Outcomes Covered

- Apply data preparation and exploratory data analysis techniques, including data summarization, data distribution analysis, outlier treatment, and measures of asymmetry, to preprocess and interpret datasets effectively.
 - Analyze datasets using descriptive statistical measures such as mean, variance, sampling methods, covariance, and correlation to identify patterns, relationships, and support data-driven decision-making.
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1. Data Summarization A dataset of 11 employee salaries (in ₹'000) is: 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120. (a) Compute the mean and median. (b) Explain why the mean is a poor measure of central tendency here. (c) Which measure of dispersion would you trust more — standard deviation or IQR — and why?

Data Summarization

Dataset of 11 employee salaries (in ₹'000): 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120.

(a) Compute the mean and median.

Sum of salaries = $32 + 35 + 36 + 38 + 40 + 42 + 45 + 48 + 50 + 95 + 120 = 581$

Mean = $\Sigma X / N = 581 / 11 = 52.82$ thousand

Since there are 11 observations, the median is the 6th value = 42 thousand.

Answer: Mean = ₹52.82 thousand; Median = ₹42 thousand.

(b) Explain why the mean is a poor measure of central tendency here.

The values 95 and 120 are unusually high compared with the other salaries. These extreme values pull the mean upward, so ₹52.82 thousand does not represent the typical employee salary as well as the median of ₹42 thousand.

Answer: The median is a more representative measure of central tendency for this dataset.

(c) Which measure of dispersion would you trust more — standard deviation or IQR — and why?

The IQR (Interquartile Range) is more reliable because it is resistant to extreme observations. Standard deviation is strongly affected by the high salaries of 95 and 120.

2. Outlier Treatment Using the dataset above, apply the $1.5 \times \text{IQR}$ rule to identify outliers. Show Q1, Q3, IQR, and the lower/upper fences. Which values qualify as outliers? If you cap them at the fences (Winsorization) instead of removing them, how would the mean change qualitatively?

Outlier Treatment

Using the $1.5 \times \text{IQR}$ rule:

Lower half = 32, 35, 36, 38, 40 $\rightarrow$ Q1 = 36

Upper half = 45, 48, 50, 95, 120 $\rightarrow$ Q3 = 50

$\text{IQR} = \text{Q3} - \text{Q1} = 50 - 36 = 14$

Lower fence = $\text{Q1} - 1.5(\text{IQR}) = 36 - 1.5(14) = 15$

Upper fence = $\text{Q3} + 1.5(\text{IQR}) = 50 + 1.5(14) = 71$

Any observation below 15 or above 71 is an outlier. Therefore, 95 and 120 qualify as outliers.

Winsorization at the upper fence

If the outliers are capped rather than removed, 95 and 120 are both replaced by 71.

New total = $581 - 95 - 120 + 71 + 71 = 508$

New mean = $508 / 11 = 46.18$ thousand

Thus, the mean decreases from 52.82 to approximately 46.18 thousand. Winsorization reduces the influence of the extreme salaries while retaining all observations.

3. Measuring Asymmetry Two datasets have the same mean (50) and standard deviation (10), but Dataset A has Pearson's skewness coefficient = +1.2 and Dataset B has -0.8. Sketch (describe) the shape of each distribution and state which is more likely to have salary-like data with a long right tail.

Measuring Asymmetry

Both datasets have Mean = 50 and Standard Deviation = 10.

Dataset A: Pearson's skewness coefficient = +1.2

A positive skewness indicates a right-skewed distribution. Most observations are concentrated toward the lower/central side, with a long tail extending toward larger values.

Shape description: a main peak toward the left/center with a long tail extending to the right.

Dataset B: Pearson's skewness coefficient = -0.8

A negative skewness indicates a left-skewed distribution, with a long tail extending toward smaller values.

Salary-like data: Dataset A is more likely because salary distributions often have a long right tail.

4. Coefficient of Skewness Calculation For a dataset: Mean = 62, Median = 58, Mode = 52, SD = 12. Calculate skewness using (a) Karl Pearson's coefficient (mode method), (b) Karl Pearson's coefficient (median method, $Sk = 3(\text{Mean} - \text{Median}) / \text{SD}$). Compare the two results and comment on why they might differ slightly.

Coefficient of Skewness Calculation

Given: Mean = 62, Median = 58, Mode = 52, Standard Deviation = 12.

(a) Karl Pearson's coefficient — Mode method

$$Sk = (\text{Mean} - \text{Mode}) / \text{SD}$$

$$Sk = (62 - 52) / 12 = 10 / 12 = 0.83$$

Therefore, skewness by the mode method = +0.83.

(b) Karl Pearson's coefficient — Median method

$$Sk = 3(\text{Mean} - \text{Median}) / \text{SD}$$

$$Sk = 3(62 - 58) / 12 = 12 / 12 = 1.00$$

Therefore, skewness by the median method = +1.00.

Method	Skewness
Mode method	0.83
Median method	1.00

Both methods indicate positive/right skewness. The results differ slightly because the formulas use different measures of central tendency (mode versus median), which can respond differently to the shape and irregularities of a distribution.

5. Variance vs Standard Deviation Interpretation Two production lines report these stats: Line A: Mean = 200 units/day, Variance = 64 Line B: Mean = 150 units/day, Variance = 36 Which line has greater relative variability? Use the Coefficient of Variation to justify your answer, not just raw variance.

Variance vs Standard Deviation Interpretation

Production Line A: Mean = 200 units/day, Variance = 64

Production Line B: Mean = 150 units/day, Variance = 36

Line A: $SD = \sqrt{64} = 8$

$CV(A) = (8 / 200) \times 100 = 4\%$

Line B: $SD = \sqrt{36} = 6$

$CV(B) = (6 / 150) \times 100 = 4\%$

Line	Mean	Variance	SD	Coefficient of Variation
A	200	64	8	4%
B	150	36	6	4%

Although Line A has the larger raw variance and standard deviation, both lines have the same coefficient of variation. Therefore, neither line has greater relative variability.

Final conclusion: $CV(A) = CV(B) = 4\%$, so both production lines have equal relative variability.